

Complete OpenFOAM case settings with y+, cores and runtime / 完整算例参数：y+、核心数、总耗时

Complete: every case now has y+, mesh quality, physical run time, actual MPI ranks and accumulated wall-clock runtime.

Runtime is accumulated from the final ClockTime in each `log.foamRun*` segment. MPI ranks are read from processor directories and OpenFOAM `nProcs` log entries. Core-hours = MPI ranks × wall-clock hours.

Summary / 总表

Case	Role	Re	Physical time	Cells	y+ min / mean / max	MPI ranks	Total wall-clock	Core-hours	Max AR	Max non-ortho	Max skew	Cd / CLrms / St
M4 Qu/Jiang	baseline low-Re validation	200	0–30.0 s	28608	0.001768 / 0.017287 / 0.032562	8	6h 58m 41s (25121 s)	55.82 core-h	117.25416	46.194952	0.93374189	1.275002 / 0.430419 / 0.187983
M5 TeacherSmooth	smooth-transition correction	200	0–30.0 s	38176	0.002314 / 0.027835 / 0.061275	8	8h 48m 44s (31724 s)	70.50 core-h	115.95065	46.445276	0.9968859	1.288671 / 0.435587 / 0.191449
M6 Young/Ooi/Krav	O-grid plus wake strip	200	0–29.999977 s	49840	0.000804 / 0.013357 / 0.02407	8	13h 30m 10s (48610 s)	108.02 core-h	130.95665	46.673541	0.88936661	1.290354 / 0.439621 / 0.191532
M7 Braza JFM	near-wake sensitivity	200	0–30.0 s	44024	0.001337 / 0.020345 / 0.03763	8	14h 0m 46s (50446 s)	112.10 core-h	125.74102	46.619023	0.91431766	1.292019 / 0.437607 / 0.192214
M8 Henderson PoF	drag/St sensitivity	200	0–30.00004 s	46960	0.000995 / 0.014617 / 0.026268	8	11h 53m 6s (42786 s)	95.08 core-h	107.53576	46.576301	1.0185337	1.295103 / 0.441585 / 0.192638
M9 Jackson JFM	wake-instability/onset	200	0–30.0 s	45600	0.001357 / 0.027789 / 0.058763	8	11h 39m 19s (41959 s)	93.24 core-h	97.868617	46.573914	0.93306564	1.288004 / 0.435875 / 0.191374
M10 Rajani AMM	fine laminar benchmark	200	0–30.000034 s	64208	0.000878 / 0.010177 / 0.019702	8	16h 2m 55s (57775 s)	128.39 core-h	124.26612	46.815189	0.92851307	1.299267 / 0.444801 / 0.193197
Andrew/Ooi Re200 Andrew/Ooi Re200	recommended display mesh	200	0–29.999973 s	49840	0.001724 / 0.010548 / 0.023868	8	11h 12m 0s (40320 s)	89.60 core-h	130.95665	46.673541	0.88936661	1.290048 / 0.437394 / 0.191530
Andrew/Ooi Re3900 Andrew/Ooi Re3900	URANS pilot, separate scope	3900	0–8.00004 s	49840	0.023465 / 0.170484 / 0.39003	8	1h 33m 38s (5618 s)	12.48 core-h	130.95665	46.673541	0.88936661	separate pilot / separate pilot / separate pilot

Runtime log evidence / 运行日志证据

Case	Log segments	Completed segments	Runtime log files
M4	1	1	log.foamRun_lit_gamg_parallel_20260912
M5	1	1	log.foamRun_lit_gamg_parallel_20260912
M6	1	1	log.foamRun_lit_gamg_parallel_20260912
M7	2	1	log.foamRun_morelit_gamg_parallel_20260912; log.foamRun_recovery_m7_braza_retry_20260915_123158
M8	2	1	log.foamRun_m8_resume_20260913_parallel_20260913_122517; log.foamRun_morelit_gamg_parallel_20260912
M9	2	1	log.foamRun_recovery_m9_resume_20260913_232917; log.foamRun_resume_queue_20260913_parallel_20260913_195602
M10	1	1	log.foamRun_recovery_m10_fresh_20260914_074310
Andrew/Ooi Re200	1	1	log.foamRun_recovery_andrew_re200_20260914_234608
Andrew/Ooi Re3900	1	1	log.foamRun_recovery_andrew_re3900_20260915_105813

M4 — Qu/Jiang

Folder: cylinderTRUE_Re200_LitMesh_QuJiang_192circ32rad_wake60_dtle04_30s_codex_20260912

Item	Symbol / setting	Value	Notes for PINN
Reynolds number	Re	200	Target Reynolds number
Cylinder diameter	D	0.001 m	Geometric/reference length scale
Free-stream velocity	U_∞	0.015 m/s	Reference velocity and inlet velocity
Kinematic viscosity	ν	7.5e-08 m ² /s	Use this value in the dimensional PINN physics loss
Flow type	—	2D, incompressible, transient, laminar	Re200 cases use laminar; Re3900 is kept separate as URANS pilot
OpenFOAM version	—	OpenFOAM 14	CFD data source
CFD solver	—	foamRun + incompressibleFluid	Incompressible transient solver
Actual parallel cores / MPI ranks	—	8	Read from processor directories and foamRun nProcs logs
Total wall-clock runtime	—	6h 58m 41s (25121 s)	Accumulated ClockTime from log.foamRun* segments
Compute cost	—	55.82 core-h	MPI ranks × wall-clock time
Runtime log source	—	log.foamRun_lit_gamg_parallel_20260912	Used to compute total runtime
Pressure–velocity coupling	—	PIMPLE	Transient pressure–velocity coupling
Cylinder centre	(xc,yc)	(0,0)	Cylinder location
Computational domain	(x,y)	approximately –0.1 to 0.1 m	Cylinder-centred domain family

Mesh size	—	28608 cells	All cells are hexahedral
Mesh dimensionality	—	2D	One cell in spanwise direction; front/back are empty/frontAndBack
Mesh quality	—	Mesh OK	Passed OpenFOAM checkMesh
Max aspect ratio	—	117.25416	Mesh-quality reference
Max non-orthogonality	—	46.194952°, average 8.1240093°	Mesh-quality reference
Max skewness	—	0.93374189	Mesh-quality reference
Cylinder y+	—	min 0.001768, mean 0.017287, max 0.032562	Computed at latest reconstructed t=30 s; samples=192
First wall-cell distance	y1	min 9.40805e-08, mean 5.76191e-07, max 6.82399e-07 m	Used in y+ calculation
Simulation time	t	0–30.0 s	Full CFD physical interval reached
Maximum CFD time step	Δtmax	0.0001 s	Maximum allowed time step
Base CFD time step	Δt	0.0001 s	controlDict deltaT
Adaptive time stepping	—	Yes	Controlled by Courant number
Maximum Courant number	Comax	0.3	CFD stability/accuracy control
PINN inputs	—	x, y, t	Neural-network inputs
PINN outputs	—	u, v, p	Predicted flow variables
x,y units	—	m	Dimensional coordinates
t unit	—	s	Physical time
u,v units	—	m/s	Velocity components
Pressure p unit	—	m ² /s ²	Kinematic pressure, NOT Pa
Inlet velocity BC	—	u=0.015, v=0	fixedValue
Outlet velocity BC	—	∂U/∂n=0	zeroGradient
Cylinder velocity BC	—	u=v=0	noSlip wall
Inlet pressure BC	—	∂p/∂n=0	zeroGradient
Outlet pressure BC	—	p=0	fixedValue pressure reference
Cylinder pressure BC	—	∂p/∂n=0	zeroGradient
PINN mapping	—	(x,y,t) → (u,v,p)	Main learning problem
Dataset file	—	M4_xyt_uvp.mat	Recommended case-specific export name
Normalisation	—	Do not assume nondimensionalised	Any normalisation must be explicitly defined in PINN code
Main validation quantities	—	CD, CL(t), f, St, u, v, p	CFD–PINN comparison
Measured Cd / CLrms / St	—	1.275002 / 0.430419 / 0.187983	Late-window Re200 force statistics; Re3900 separate pilot

M5 — TeacherSmooth

Folder: cylinderTRUE_Re200_LitMesh_TeacherSmooth_224circ36rad_wake70_dt1e04_30s_codex_20260912

Item	Symbol / setting	Value	Notes for PINN
Reynolds number	Re	200	Target Reynolds number
Cylinder diameter	D	0.001 m	Geometric/reference length scale
Free-stream velocity	U _∞	0.015 m/s	Reference velocity and inlet velocity
Kinematic viscosity	ν	7.5e-08 m ² /s	Use this value in the dimensional PINN physics loss
Flow type	—	2D, incompressible, transient, laminar	Re200 cases use laminar; Re3900 is kept separate as URANS pilot
OpenFOAM version	—	OpenFOAM 14	CFD data source
CFD solver	—	foamRun + incompressibleFluid	Incompressible transient solver
Actual parallel cores / MPI ranks	—	8	Read from processor directories and foamRun nProcs logs
Total wall-clock runtime	—	8h 48m 44s (31724 s)	Accumulated ClockTime from log.foamRun* segments
Compute cost	—	70.50 core-h	MPI ranks × wall-clock time
Runtime log source	—	log.foamRun_lit_gamg_parallel_20260912	Used to compute total runtime
Pressure–velocity coupling	—	PIMPLE	Transient pressure–velocity coupling
Cylinder centre	(xc,yc)	(0,0)	Cylinder location
Computational domain	(x,y)	approximately –0.1 to 0.1 m	Cylinder-centred domain family
Mesh size	—	38176 cells	All cells are hexahedral
Mesh dimensionality	—	2D	One cell in spanwise direction; front/back are empty/frontAndBack
Mesh quality	—	Mesh OK	Passed OpenFOAM checkMesh
Max aspect ratio	—	115.95065	Mesh-quality reference
Max non-orthogonality	—	46.445276°, average 7.8302268°	Mesh-quality reference
Max skewness	—	0.9968859	Mesh-quality reference
Cylinder y+	—	min 0.002314, mean 0.027835, max 0.061275	Computed at latest reconstructed t=30 s; samples=224
First wall-cell distance	y1	min 5.48923e-07, mean 9.34365e-07, max 1.01294e-06 m	Used in y+ calculation
Simulation time	t	0–30.0 s	Full CFD physical interval reached
Maximum CFD time step	Δtmax	0.0001 s	Maximum allowed time step
Base CFD time step	Δt	0.0001 s	controlDict deltaT
Adaptive time stepping	—	Yes	Controlled by Courant number

Maximum Courant number	Comax	0.3	CFD stability/accuracy control
PINN inputs	—	x, y, t	Neural-network inputs
PINN outputs	—	u, v, p	Predicted flow variables
x,y units	—	m	Dimensional coordinates
t unit	—	s	Physical time
u,v units	—	m/s	Velocity components
Pressure p unit	—	m ² /s ²	Kinematic pressure, NOT Pa
Inlet velocity BC	—	u=0.015, v=0	fixedValue
Outlet velocity BC	—	$\partial U/\partial n=0$	zeroGradient
Cylinder velocity BC	—	u=v=0	noSlip wall
Inlet pressure BC	—	$\partial p/\partial n=0$	zeroGradient
Outlet pressure BC	—	p=0	fixedValue pressure reference
Cylinder pressure BC	—	$\partial p/\partial n=0$	zeroGradient
PINN mapping	—	(x,y,t) $\rightarrow$ (u,v,p)	Main learning problem
Dataset file	—	M5_xyt_uvp.mat	Recommended case-specific export name
Normalisation	—	Do not assume nondimensionalised	Any normalisation must be explicitly defined in PINN code
Main validation quantities	—	CD, CL(t), ξ , St, u, v, p	CFD–PINN comparison
Measured Cd / CLrms / St	—	1.288671 / 0.435587 / 0.191449	Late-window Re200 force statistics; Re3900 separate pilot

M6 — Young/Ooi/Krav

Folder: cylinderTRUE_Re200_LitMesh_YoungOoiKrav_256circ40rad_wake80_dtle04_30s_codex_20260912

Item	Symbol / setting	Value	Notes for PINN
Reynolds number	Re	200	Target Reynolds number
Cylinder diameter	D	0.001 m	Geometric/reference length scale
Free-stream velocity	U ∞	0.015 m/s	Reference velocity and inlet velocity
Kinematic viscosity	ν	7.5e-08 m ² /s	Use this value in the dimensional PINN physics loss
Flow type	—	2D, incompressible, transient, laminar	Re200 cases use laminar; Re3900 is kept separate as URANS pilot
OpenFOAM version	—	OpenFOAM 14	CFD data source
CFD solver	—	foamRun + incompressibleFluid	Incompressible transient solver
Actual parallel cores / MPI ranks	—	8	Read from processor directories and foamRun nProcs logs
Total wall-clock runtime	—	13h 30m 10s (48610 s)	Accumulated ClockTime from log.foamRun* segments
Compute cost	—	108.02 core-h	MPI ranks $\times$ wall-clock time
Runtime log source	—	log.foamRun_lit_gamg_parallel_20260912	Used to compute total runtime
Pressure–velocity coupling	—	PIMPLE	Transient pressure–velocity coupling
Cylinder centre	(xc,yc)	(0,0)	Cylinder location
Computational domain	(x,y)	approximately -0.1 to 0.1 m	Cylinder-centred domain family
Mesh size	—	49840 cells	All cells are hexahedral
Mesh dimensionality	—	2D	One cell in spanwise direction; front/back are empty/frontAndBack
Mesh quality	—	Mesh OK	Passed OpenFOAM checkMesh
Max aspect ratio	—	130.95665	Mesh-quality reference
Max non-orthogonality	—	46.673541°, average 7.989432°	Mesh-quality reference
Max skewness	—	0.88936661	Mesh-quality reference
Cylinder y+	—	min 0.000804, mean 0.013357, max 0.02407	Computed at latest reconstructed t=29.999977 s; samples=256
First wall-cell distance	y1	min 1.30726e-07, mean 4.44997e-07, max 5.07911e-07 m	Used in y+ calculation
Simulation time	t	0–29.999977 s	Full CFD physical interval reached
Maximum CFD time step	Δt_{max}	0.0001 s	Maximum allowed time step
Base CFD time step	Δt	0.0001 s	controlDict deltaT
Adaptive time stepping	—	Yes	Controlled by Courant number
Maximum Courant number	Comax	0.3	CFD stability/accuracy control
PINN inputs	—	x, y, t	Neural-network inputs
PINN outputs	—	u, v, p	Predicted flow variables
x,y units	—	m	Dimensional coordinates
t unit	—	s	Physical time
u,v units	—	m/s	Velocity components
Pressure p unit	—	m ² /s ²	Kinematic pressure, NOT Pa
Inlet velocity BC	—	u=0.015, v=0	fixedValue
Outlet velocity BC	—	$\partial U/\partial n=0$	zeroGradient
Cylinder velocity BC	—	u=v=0	noSlip wall
Inlet pressure BC	—	$\partial p/\partial n=0$	zeroGradient
Outlet pressure BC	—	p=0	fixedValue pressure reference

Cylinder pressure BC	—	$\partial p / \partial n = 0$	zeroGradient
PINN mapping	—	$(x, y, t) \rightarrow (u, v, p)$	Main learning problem
Dataset file	—	M6_xyt_uvp.mat	Recommended case-specific export name
Normalisation	—	Do not assume nondimensionalised	Any normalisation must be explicitly defined in PINN code
Main validation quantities	—	CD, CL(t), ξ St, u, v, p	CFD–PINN comparison
Measured Cd / CLrms / St	—	1.290354 / 0.439621 / 0.191532	Late-window Re200 force statistics; Re3900 separate pilot

M7 — Braza JFM

Folder: cylinderTRUE_Re200_LitMesh_BrazaJFM_nearWake_224circ40rad_wake90_dtle04_30s_codex_20260912

Item	Symbol / setting	Value	Notes for PINN
Reynolds number	Re	200	Target Reynolds number
Cylinder diameter	D	0.001 m	Geometric/reference length scale
Free-stream velocity	U_∞	0.015 m/s	Reference velocity and inlet velocity
Kinematic viscosity	ν	7.5e-08 m ² /s	Use this value in the dimensional PINN physics loss
Flow type	—	2D, incompressible, transient, laminar	Re200 cases use laminar; Re3900 is kept separate as URANS pilot
OpenFOAM version	—	OpenFOAM 14	CFD data source
CFD solver	—	foamRun + incompressibleFluid	Incompressible transient solver
Actual parallel cores / MPI ranks	—	8	Read from processor directories and foamRun nProcs logs
Total wall-clock runtime	—	14h 0m 46s (50446 s)	Accumulated ClockTime from log.foamRun* segments
Compute cost	—	112.10 core-h	MPI ranks × wall-clock time
Runtime log source	—	log.foamRun_morelit_gamg_parallel_20260912; log.foamRun_recovery_m7_braza_retry_20260915_123158	Used to compute total runtime
Pressure–velocity coupling	—	PIMPLE	Transient pressure–velocity coupling
Cylinder centre	(xc,yc)	(0,0)	Cylinder location
Computational domain	(x,y)	approximately −0.1 to 0.1 m	Cylinder-centred domain family
Mesh size	—	44024 cells	All cells are hexahedral
Mesh dimensionality	—	2D	One cell in spanwise direction; front/back are empty/frontAndBack
Mesh quality	—	Mesh OK	Passed OpenFOAM checkMesh
Max aspect ratio	—	125.74102	Mesh-quality reference
Max non-orthogonality	—	46.619023°, average 8.5305483°	Mesh-quality reference
Max skewness	—	0.91431766	Mesh-quality reference
Cylinder y+	—	min 0.001337, mean 0.020345, max 0.03763	Computed at latest reconstructed t=30 s; samples=224
First wall-cell distance	y1	min 2.77683e-07, mean 6.62916e-07, max 7.41448e-07 m	Used in y+ calculation
Simulation time	t	0–30.0 s	Full CFD physical interval reached
Maximum CFD time step	$\Delta t_{\max}$	0.0001 s	Maximum allowed time step
Base CFD time step	Δt	0.0001 s	controlDict deltaT
Adaptive time stepping	—	Yes	Controlled by Courant number
Maximum Courant number	Comax	0.3	CFD stability/accuracy control
PINN inputs	—	x, y, t	Neural-network inputs
PINN outputs	—	u, v, p	Predicted flow variables
x,y units	—	m	Dimensional coordinates
t unit	—	s	Physical time
u,v units	—	m/s	Velocity components
Pressure p unit	—	m ² /s ²	Kinematic pressure, NOT Pa
Inlet velocity BC	—	u=0.015, v=0	fixedValue
Outlet velocity BC	—	$\partial U / \partial n = 0$	zeroGradient
Cylinder velocity BC	—	u=v=0	noSlip wall
Inlet pressure BC	—	$\partial p / \partial n = 0$	zeroGradient
Outlet pressure BC	—	p=0	fixedValue pressure reference
Cylinder pressure BC	—	$\partial p / \partial n = 0$	zeroGradient
PINN mapping	—	$(x, y, t) \rightarrow (u, v, p)$	Main learning problem
Dataset file	—	M7_xyt_uvp.mat	Recommended case-specific export name
Normalisation	—	Do not assume nondimensionalised	Any normalisation must be explicitly defined in PINN code
Main validation quantities	—	CD, CL(t), ξ St, u, v, p	CFD–PINN comparison
Measured Cd / CLrms / St	—	1.292019 / 0.437607 / 0.192214	Late-window Re200 force statistics; Re3900 separate pilot

M8 — Henderson PoF

Folder: cylinderTRUE_Re200_LitMesh_HendersonPoF_drag_256circ44rad_wake72_dtle04_30s_codex_20260912

Item	Symbol / setting	Value	Notes for PINN
Reynolds number	Re	200	Target Reynolds number
Cylinder diameter	D	0.001 m	Geometric/reference length scale

Free-stream velocity	U_∞	0.015 m/s	Reference velocity and inlet velocity
Kinematic viscosity	ν	7.5e-08 m ² /s	Use this value in the dimensional PINN physics loss
Flow type	—	2D, incompressible, transient, laminar	Re200 cases use laminar; Re3900 is kept separate as URANS pilot
OpenFOAM version	—	OpenFOAM 14	CFD data source
CFD solver	—	foamRun + incompressibleFluid	Incompressible transient solver
Actual parallel cores / MPI ranks	—	8	Read from processor directories and foamRun nProcs logs
Total wall-clock runtime	—	11h 53m 6s (42786 s)	Accumulated ClockTime from log.foamRun* segments
Compute cost	—	95.08 core-h	MPI ranks × wall-clock time
Runtime log source	—	log.foamRun_m8_resume_20260913_parallel_20260913_122517; log.foamRun_morelit_gamg_parallel_20260912	Used to compute total runtime
Pressure–velocity coupling	—	PIMPLE	Transient pressure–velocity coupling
Cylinder centre	(xc,yc)	(0,0)	Cylinder location
Computational domain	(x,y)	approximately −0.1 to 0.1 m	Cylinder-centred domain family
Mesh size	—	46960 cells	All cells are hexahedral
Mesh dimensionality	—	2D	One cell in spanwise direction; front/back are empty/frontAndBack
Mesh quality	—	Mesh OK	Passed OpenFOAM checkMesh
Max aspect ratio	—	107.53576	Mesh-quality reference
Max non-orthogonality	—	46.576301°, average 7.7292209°	Mesh-quality reference
Max skewness	—	1.0185337	Mesh-quality reference
Cylinder y+	—	min 0.000995, mean 0.014617, max 0.026268	Computed at latest reconstructed t=30.00004 s; samples=256
First wall-cell distance	y1	min 1.6147e-07, mean 4.75806e-07, max 5.38014e-07 m	Used in y+ calculation
Simulation time	t	0–30.00004 s	Full CFD physical interval reached
Maximum CFD time step	$\Delta t_{\max}$	0.0001 s	Maximum allowed time step
Base CFD time step	Δt	0.0001 s	controlDict deltaT
Adaptive time stepping	—	Yes	Controlled by Courant number
Maximum Courant number	Comax	0.3	CFD stability/accuracy control
PINN inputs	—	x, y, t	Neural-network inputs
PINN outputs	—	u, v, p	Predicted flow variables
x,y units	—	m	Dimensional coordinates
t unit	—	s	Physical time
u,v units	—	m/s	Velocity components
Pressure p unit	—	m ² /s ²	Kinematic pressure, NOT Pa
Inlet velocity BC	—	u=0.015, v=0	fixedValue
Outlet velocity BC	—	$\partial U / \partial n = 0$	zeroGradient
Cylinder velocity BC	—	u=v=0	noSlip wall
Inlet pressure BC	—	$\partial p / \partial n = 0$	zeroGradient
Outlet pressure BC	—	p=0	fixedValue pressure reference
Cylinder pressure BC	—	$\partial p / \partial n = 0$	zeroGradient
PINN mapping	—	(x,y,t) → (u,v,p)	Main learning problem
Dataset file	—	M8_xyt_uvp.mat	Recommended case-specific export name
Normalisation	—	Do not assume nondimensionalised	Any normalisation must be explicitly defined in PINN code
Main validation quantities	—	CD, CL(t), f, St, u, v, p	CFD–PINN comparison
Measured Cd / CLrms / St	—	1.295103 / 0.441585 / 0.192638	Late-window Re200 force statistics; Re3900 separate pilot

M9 — Jackson JFM

Folder: cylinderTRUE_Re200_LitMesh_JacksonJFM_onset_224circ36rad_smooth_dtle04_30s_codex_20260912

Item	Symbol / setting	Value	Notes for PINN
Reynolds number	Re	200	Target Reynolds number
Cylinder diameter	D	0.001 m	Geometric/reference length scale
Free-stream velocity	U_∞	0.015 m/s	Reference velocity and inlet velocity
Kinematic viscosity	ν	7.5e-08 m ² /s	Use this value in the dimensional PINN physics loss
Flow type	—	2D, incompressible, transient, laminar	Re200 cases use laminar; Re3900 is kept separate as URANS pilot
OpenFOAM version	—	OpenFOAM 14	CFD data source
CFD solver	—	foamRun + incompressibleFluid	Incompressible transient solver
Actual parallel cores / MPI ranks	—	8	Read from processor directories and foamRun nProcs logs
Total wall-clock runtime	—	11h 39m 19s (41959 s)	Accumulated ClockTime from log.foamRun* segments
Compute cost	—	93.24 core-h	MPI ranks × wall-clock time
Runtime log source	—	log.foamRun_recovery_m9_resume_20260913_232917; log.foamRun_resume_queue_20260913_parallel_20260913_195602	Used to compute total runtime
Pressure–velocity coupling	—	PIMPLE	Transient pressure–velocity coupling
Cylinder centre	(xc,yc)	(0,0)	Cylinder location

Computational domain	(x,y)	approximately -0.1 to 0.1 m	Cylinder-centred domain family
Mesh size	—	45600 cells	All cells are hexahedral
Mesh dimensionality	—	2D	One cell in spanwise direction; front/back are empty/frontAndBack
Mesh quality	—	Mesh OK	Passed OpenFOAM checkMesh
Max aspect ratio	—	97.868617	Mesh-quality reference
Max non-orthogonality	—	46.573914°, average 8.3280936°	Mesh-quality reference
Max skewness	—	0.93306564	Mesh-quality reference
Cylinder y+	—	min 0.001357, mean 0.027789, max 0.058763	Computed at latest reconstructed t=30 s; samples=224
First wall-cell distance	y1	min 5.1581e-07, mean 9.01276e-07, max 9.7872e-07 m	Used in y+ calculation
Simulation time	t	0–30.0 s	Full CFD physical interval reached
Maximum CFD time step	Δtmax	0.0001 s	Maximum allowed time step
Base CFD time step	Δt	0.0001 s	controlDict deltaT
Adaptive time stepping	—	Yes	Controlled by Courant number
Maximum Courant number	Comax	0.3	CFD stability/accuracy control
PINN inputs	—	x, y, t	Neural-network inputs
PINN outputs	—	u, v, p	Predicted flow variables
x,y units	—	m	Dimensional coordinates
t unit	—	s	Physical time
u,v units	—	m/s	Velocity components
Pressure p unit	—	m ² /s ²	Kinematic pressure, NOT Pa
Inlet velocity BC	—	u=0.015, v=0	fixedValue
Outlet velocity BC	—	∂U/∂n=0	zeroGradient
Cylinder velocity BC	—	u=v=0	noSlip wall
Inlet pressure BC	—	∂p/∂n=0	zeroGradient
Outlet pressure BC	—	p=0	fixedValue pressure reference
Cylinder pressure BC	—	∂p/∂n=0	zeroGradient
PINN mapping	—	(x,y,t) → (u,v,p)	Main learning problem
Dataset file	—	M9_xy_t_uvp.mat	Recommended case-specific export name
Normalisation	—	Do not assume nondimensionalised	Any normalisation must be explicitly defined in PINN code
Main validation quantities	—	CD, CL(t), f, St, u, v, p	CFD–PINN comparison
Measured Cd / CLrms / St	—	1.288004 / 0.435875 / 0.191374	Late-window Re200 force statistics; Re3900 separate pilot

M10 — Rajani AMM

Folder: cylinderTRUE_Re200_LitMesh_RajaniAMM_laminar_288circ48rad_wake90_dtle04_30s_codex_20260912

Item	Symbol / setting	Value	Notes for PINN
Reynolds number	Re	200	Target Reynolds number
Cylinder diameter	D	0.001 m	Geometric/reference length scale
Free-stream velocity	U _∞	0.015 m/s	Reference velocity and inlet velocity
Kinematic viscosity	ν	7.5e-08 m ² /s	Use this value in the dimensional PINN physics loss
Flow type	—	2D, incompressible, transient, laminar	Re200 cases use laminar; Re3900 is kept separate as URANS pilot
OpenFOAM version	—	OpenFOAM 14	CFD data source
CFD solver	—	foamRun + incompressibleFluid	Incompressible transient solver
Actual parallel cores / MPI ranks	—	8	Read from processor directories and foamRun nProcs logs
Total wall-clock runtime	—	16h 2m 55s (57775 s)	Accumulated ClockTime from log.foamRun* segments
Compute cost	—	128.39 core-h	MPI ranks × wall-clock time
Runtime log source	—	log.foamRun_recovery_m10_fresh_20260914_074310	Used to compute total runtime
Pressure–velocity coupling	—	PIMPLE	Transient pressure–velocity coupling
Cylinder centre	(xc,yc)	(0,0)	Cylinder location
Computational domain	(x,y)	approximately -0.1 to 0.1 m	Cylinder-centred domain family
Mesh size	—	64208 cells	All cells are hexahedral
Mesh dimensionality	—	2D	One cell in spanwise direction; front/back are empty/frontAndBack
Mesh quality	—	Mesh OK	Passed OpenFOAM checkMesh
Max aspect ratio	—	124.26612	Mesh-quality reference
Max non-orthogonality	—	46.815189°, average 7.9148679°	Mesh-quality reference
Max skewness	—	0.92851307	Mesh-quality reference
Cylinder y+	—	min 0.000878, mean 0.01077, max 0.019702	Computed at latest reconstructed t=30.000034 s; samples=288
First wall-cell distance	y1	min 1.15516e-07, mean 3.76782e-07, max 4.27319e-07 m	Used in y+ calculation
Simulation time	t	0–30.000034 s	Full CFD physical interval reached
Maximum CFD time step	Δtmax	0.0001 s	Maximum allowed time step
Base CFD time step	Δt	0.0001 s	controlDict deltaT

Adaptive time stepping	—	Yes	Controlled by Courant number
Maximum Courant number	Comax	0.3	CFD stability/accuracy control
PINN inputs	—	x, y, t	Neural-network inputs
PINN outputs	—	u, v, p	Predicted flow variables
x, y units	—	m	Dimensional coordinates
t unit	—	s	Physical time
u, v units	—	m/s	Velocity components
Pressure p unit	—	m^2/s^2	Kinematic pressure, NOT Pa
Inlet velocity BC	—	$u=0.015, v=0$	fixedValue
Outlet velocity BC	—	$\partial U/\partial n=0$	zeroGradient
Cylinder velocity BC	—	$u=v=0$	noSlip wall
Inlet pressure BC	—	$\partial p/\partial n=0$	zeroGradient
Outlet pressure BC	—	$p=0$	fixedValue pressure reference
Cylinder pressure BC	—	$\partial p/\partial n=0$	zeroGradient
PINN mapping	—	$(x, y, t) \rightarrow (u, v, p)$	Main learning problem
Dataset file	—	M10_xyt_uvp.mat	Recommended case-specific export name
Normalisation	—	Do not assume nondimensionalised	Any normalisation must be explicitly defined in PINN code
Main validation quantities	—	$CD, CL(t), f, St, u, v, p$	CFD–PINN comparison
Measured $Cd / CLrms / St$	—	1.299267 / 0.444801 / 0.193197	Late-window Re200 force statistics; Re3900 separate pilot

Andrew/Ooi Re200 — Andrew/Ooi Re200

Folder: cylinderTRUE_Re200_AndrewOoiGrid_laminar_dtle04_30s_codex_20260912

Item	Symbol / setting	Value	Notes for PINN
Reynolds number	Re	200	Target Reynolds number
Cylinder diameter	D	0.001 m	Geometric/reference length scale
Free-stream velocity	U_∞	0.015 m/s	Reference velocity and inlet velocity
Kinematic viscosity	ν	7.5e-08 m^2/s	Use this value in the dimensional PINN physics loss
Flow type	—	2D, incompressible, transient, laminar	Re200 cases use laminar; Re3900 is kept separate as URANS pilot
OpenFOAM version	—	OpenFOAM 14	CFD data source
CFD solver	—	foamRun + incompressibleFluid	Incompressible transient solver
Actual parallel cores / MPI ranks	—	8	Read from processor directories and foamRun nProcs logs
Total wall-clock runtime	—	11h 12m 0s (40320 s)	Accumulated ClockTime from log.foamRun* segments
Compute cost	—	89.60 core-h	MPI ranks × wall-clock time
Runtime log source	—	log.foamRun_recovery_andrew_re200_20260914_234608	Used to compute total runtime
Pressure–velocity coupling	—	PIMPLE	Transient pressure–velocity coupling
Cylinder centre	(xc,yc)	(0,0)	Cylinder location
Computational domain	(x,y)	approximately −0.1 to 0.1 m	Cylinder-centred domain family
Mesh size	—	49840 cells	All cells are hexahedral
Mesh dimensionality	—	2D	One cell in spanwise direction; front/back are empty/frontAndBack
Mesh quality	—	Mesh OK	Passed OpenFOAM checkMesh
Max aspect ratio	—	130.95665	Mesh-quality reference
Max non-orthogonality	—	46.673541°, average 7.9894327°	Mesh-quality reference
Max skewness	—	0.88936661	Mesh-quality reference
Cylinder y^+	—	min 0.001724, mean 0.010548, max 0.023868	Computed at latest reconstructed $t=29.999973$ s; samples=256
First wall-cell distance	y_1	min 1.30726e-07, mean 4.44997e-07, max 5.07911e-07 m	Used in y^+ calculation
Simulation time	t	0–29.999973 s	Full CFD physical interval reached
Maximum CFD time step	Δt_{max}	0.0001 s	Maximum allowed time step
Base CFD time step	Δt	0.0001 s	controlDict deltaT
Adaptive time stepping	—	Yes	Controlled by Courant number
Maximum Courant number	Comax	0.3	CFD stability/accuracy control
PINN inputs	—	x, y, t	Neural-network inputs
PINN outputs	—	u, v, p	Predicted flow variables
x, y units	—	m	Dimensional coordinates
t unit	—	s	Physical time
u, v units	—	m/s	Velocity components
Pressure p unit	—	m^2/s^2	Kinematic pressure, NOT Pa
Inlet velocity BC	—	$u=0.015, v=0$	fixedValue
Outlet velocity BC	—	$\partial U/\partial n=0$	zeroGradient
Cylinder velocity BC	—	$u=v=0$	noSlip wall
Inlet pressure BC	—	$\partial p/\partial n=0$	zeroGradient

Outlet pressure BC	—	$p=0$	fixedValue pressure reference
Cylinder pressure BC	—	$\partial p/\partial n=0$	zeroGradient
PINN mapping	—	$(x,y,t) \rightarrow (u,v,p)$	Main learning problem
Dataset file	—	Andrew_Ooi_Re200_xyt_uvp.mat	Recommended case-specific export name
Normalisation	—	Do not assume nondimensionalised	Any normalisation must be explicitly defined in PINN code
Main validation quantities	—	CD, CL(t), $\bar{f}$, St, u, v, p	CFD–PINN comparison
Measured Cd / CLrms / St	—	1.290048 / 0.437394 / 0.191530	Late-window Re200 force statistics; Re3900 separate pilot

Andrew/Ooi Re3900 — Andrew/Ooi Re3900

Folder: cylinderTRUE_Re3900_AndrewOoiGrid_URANS_SST_dt2e04_8s_codex_20260912

Item	Symbol / setting	Value	Notes for PINN
Reynolds number	Re	3900	Target Reynolds number
Cylinder diameter	D	0.001 m	Geometric/reference length scale
Free-stream velocity	U_∞	0.015 m/s	Reference velocity and inlet velocity
Kinematic viscosity	ν	3.8461538e-09 m ² /s	Use this value in the dimensional PINN physics loss
Flow type	—	2D, incompressible, transient, URANS/RAS SST pilot	Re200 cases use laminar; Re3900 is kept separate as URANS pilot
OpenFOAM version	—	OpenFOAM 14	CFD data source
CFD solver	—	foamRun + incompressibleFluid	Incompressible transient solver
Actual parallel cores / MPI ranks	—	8	Read from processor directories and foamRun nProcs logs
Total wall-clock runtime	—	1h 33m 38s (5618 s)	Accumulated ClockTime from log.foamRun* segments
Compute cost	—	12.48 core-h	MPI ranks × wall-clock time
Runtime log source	—	log.foamRun_recovery_andrew_re3900_20260915_105813	Used to compute total runtime
Pressure–velocity coupling	—	PIMPLE	Transient pressure–velocity coupling
Cylinder centre	(xc,yc)	(0,0)	Cylinder location
Computational domain	(x,y)	approximately –0.1 to 0.1 m	Cylinder-centred domain family
Mesh size	—	49840 cells	All cells are hexahedral
Mesh dimensionality	—	2D	One cell in spanwise direction; front/back are empty/frontAndBack
Mesh quality	—	Mesh OK	Passed OpenFOAM checkMesh
Max aspect ratio	—	130.95665	Mesh-quality reference
Max non-orthogonality	—	46.673541°, average 7.9894327°	Mesh-quality reference
Max skewness	—	0.88936661	Mesh-quality reference
Cylinder y+	—	min 0.023465, mean 0.170484, max 0.39003	Computed at latest reconstructed t=8.00004 s; samples=256
First wall-cell distance	y1	min 1.30726e-07, mean 4.44997e-07, max 5.07911e-07 m	Used in y+ calculation
Simulation time	t	0–8.00004 s	Full CFD physical interval reached
Maximum CFD time step	$\Delta t_{\max}$	0.0002 s	Maximum allowed time step
Base CFD time step	Δt	0.0002 s	controlDict deltaT
Adaptive time stepping	—	Yes	Controlled by Courant number
Maximum Courant number	Comax	0.5	CFD stability/accuracy control
PINN inputs	—	x, y, t	Neural-network inputs
PINN outputs	—	u, v, p	Predicted flow variables
x,y units	—	m	Dimensional coordinates
t unit	—	s	Physical time
u,v units	—	m/s	Velocity components
Pressure p unit	—	m ² /s ²	Kinematic pressure, NOT Pa
Inlet velocity BC	—	u=0.015, v=0	fixedValue
Outlet velocity BC	—	$\partial U/\partial n=0$	zeroGradient
Cylinder velocity BC	—	u=v=0	noSlip wall
Inlet pressure BC	—	$\partial p/\partial n=0$	zeroGradient
Outlet pressure BC	—	$p=0$	fixedValue pressure reference
Cylinder pressure BC	—	$\partial p/\partial n=0$	zeroGradient
PINN mapping	—	$(x,y,t) \rightarrow (u,v,p)$	Main learning problem
Dataset file	—	Andrew_Ooi_Re3900_xyt_uvp.mat	Recommended case-specific export name
Normalisation	—	Do not assume nondimensionalised	Any normalisation must be explicitly defined in PINN code
Main validation quantities	—	CD, CL(t), $\bar{f}$, St, u, v, p	CFD–PINN comparison
Measured Cd / CLrms / St	—	separate pilot / separate pilot / separate pilot	Late-window Re200 force statistics; Re3900 separate pilot